



Chapter 3

Manipulator Forward Kinematics

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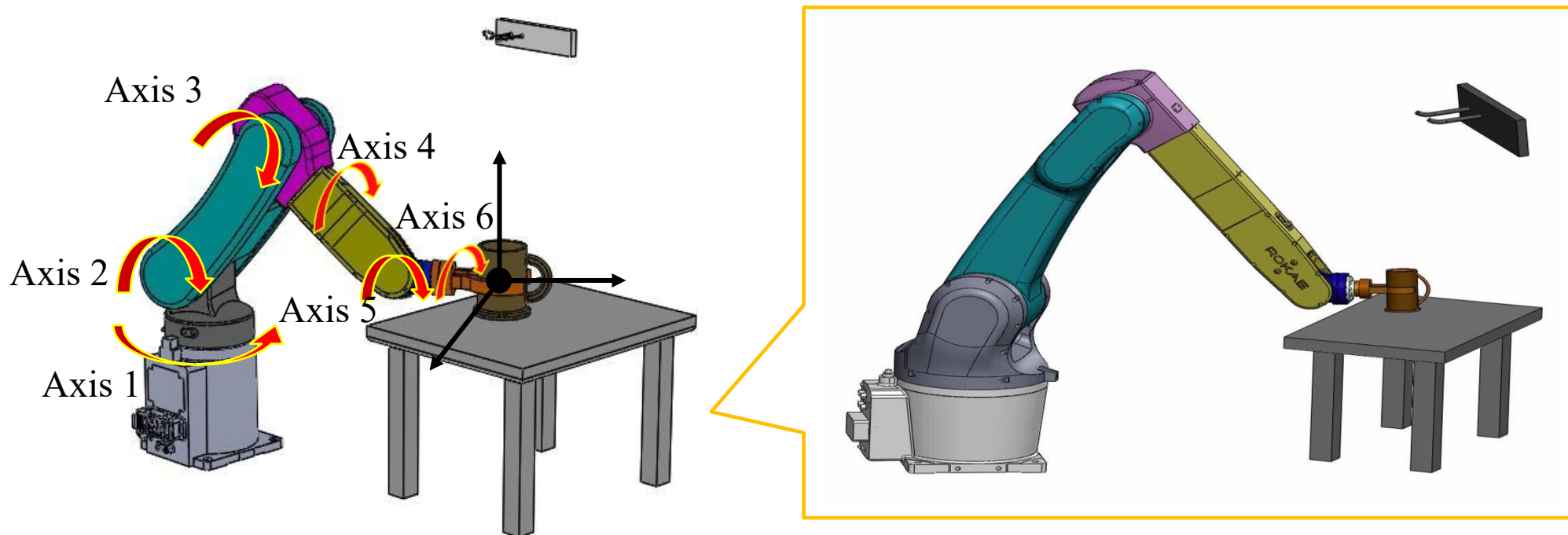


Chapter 3 Manipulator Forward Kinematics

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3. Definition of the Frame of Link
4. Manipulator Forward Kinematics
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Questions to Consider

- ❑ How to achieve a specific motion of the manipulator's end-effector?
- ❑ What is the difference between the motion of the manipulator's end-effector and that of the individual joints?
- ❑ If the joint angles of the manipulator are known, how can the position and orientation of the end-effector be determined?





1. Introduction

□ **Kinematics:** Discusses the motion state without regard to the "force" that cause it.

➤ **Position (x), velocity (v), acceleration (a), and time (t)** relationships

➤ Translation/Rotation

- Position/Posture
- Velocity/Angular Velocity
- Acceleration/Angular Acceleration

$$v = \frac{d}{dt} x$$

$$a = \frac{d}{dt} v \quad a = \frac{d^2}{dt^2} x \quad v dv = a dx$$

All the geometrical and time-based properties of the motion

□ **Dynamics :** Discuss how force/torque produces motion

➤ Newton's Second Law

$$\sum F = ma$$

➤ Work & Energy

$$T_1 + V_1 + U_{1-2}' = T_2 + V_2$$

➤ Impulse & Momentum

$$\int \sum F dt = G_2 - G_1$$

1. Introduction

□ Robotic Manipulator

- Consists of multiple links connected in series, with complex geometric shapes.
- The relative motion between links (prismatic or revolute) is driven by actuators.

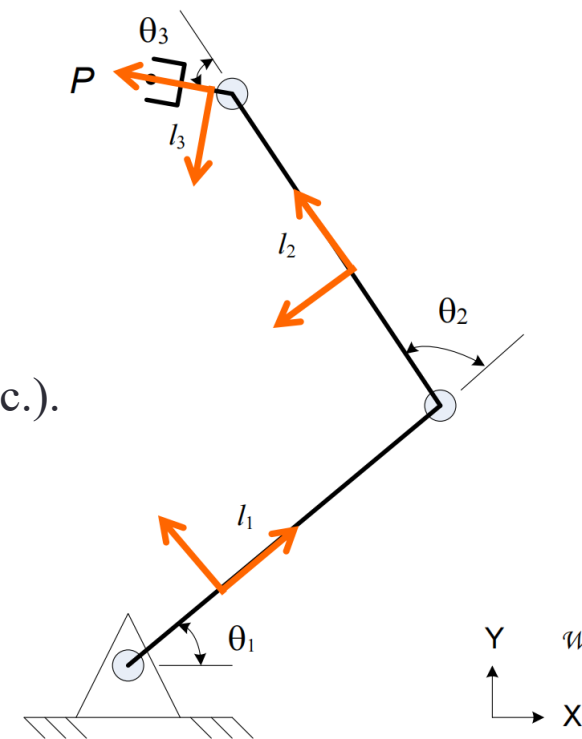
□ Problem

- **Requirement:** the kinematic state of the end-effector (e.g., position wP , velocity, etc.).
- **How to Achieve:** by driving the actuator of each joint

$${}^wP = f(\theta_1, \theta_2, \dots, \theta_n)$$

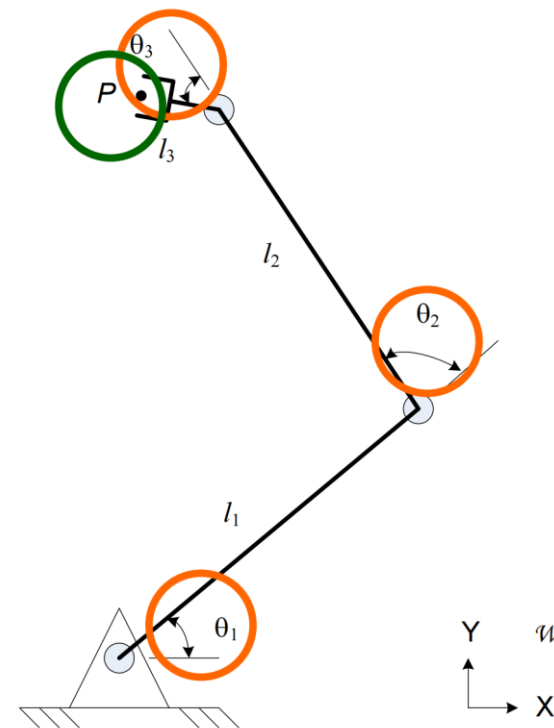
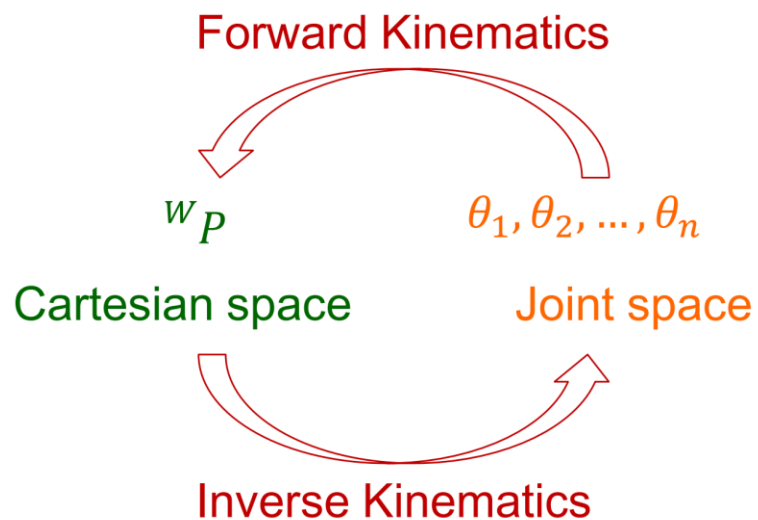
□ Method to Describe the State of a Manipulator

- Identify the relative configurations between the links.
- Establish a frame on each link, using the frame's state to represent the link's configuration.



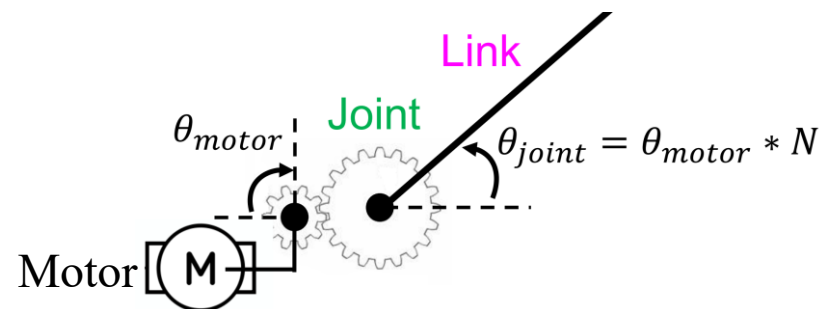
1. Introduction

□ Joint space and Cartesian space



□ Actuator space and joint space

- Determined by the mechanism that connects the actuator to the joint.



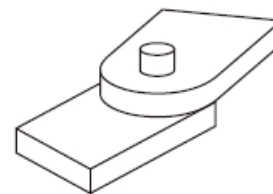
Reducer, gear ratio (N) = 1/3



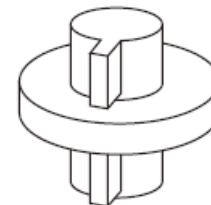
2. Geometric Description of the Manipulator

□ Joint

- Each revolute or prismatic joint has 1 Degree of Freedom (DOF).
- Each joint performs rotation or translation around a **specific axis**.



Revolute

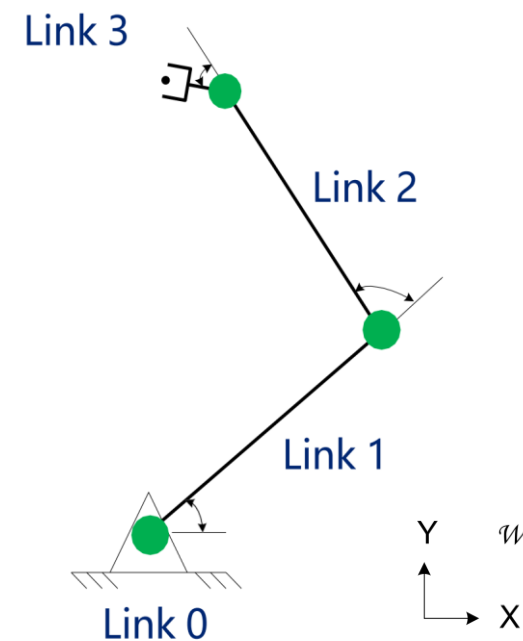


Prismatic

□ Link

- Links connect joints and are rigid bodies.
- Numbering method:
 - Link 0: The fixed base
 - Link 1: Connected to Link 0, the first moving body
 - Link 2: The second moving body
 - ...

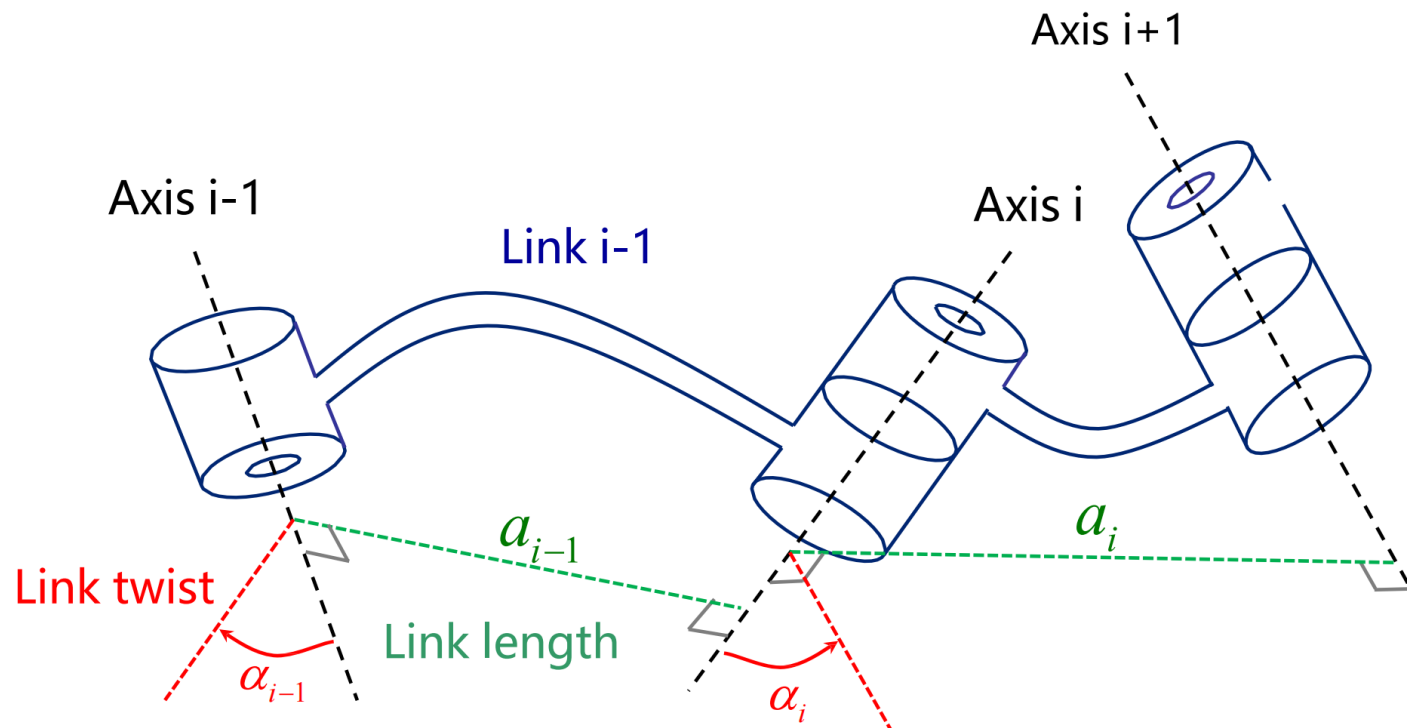
□ Typical manipulators have five or six joints





2. Geometric Description of the Manipulator

- ❑ In 3D space, for any **two arbitrary axes**, there is a **mutually perpendicular line** that is perpendicular to both axes.
- ❑ For a given links, parameters a and α can be used to clearly define the relationship.



Link length a : the distant between two axes.

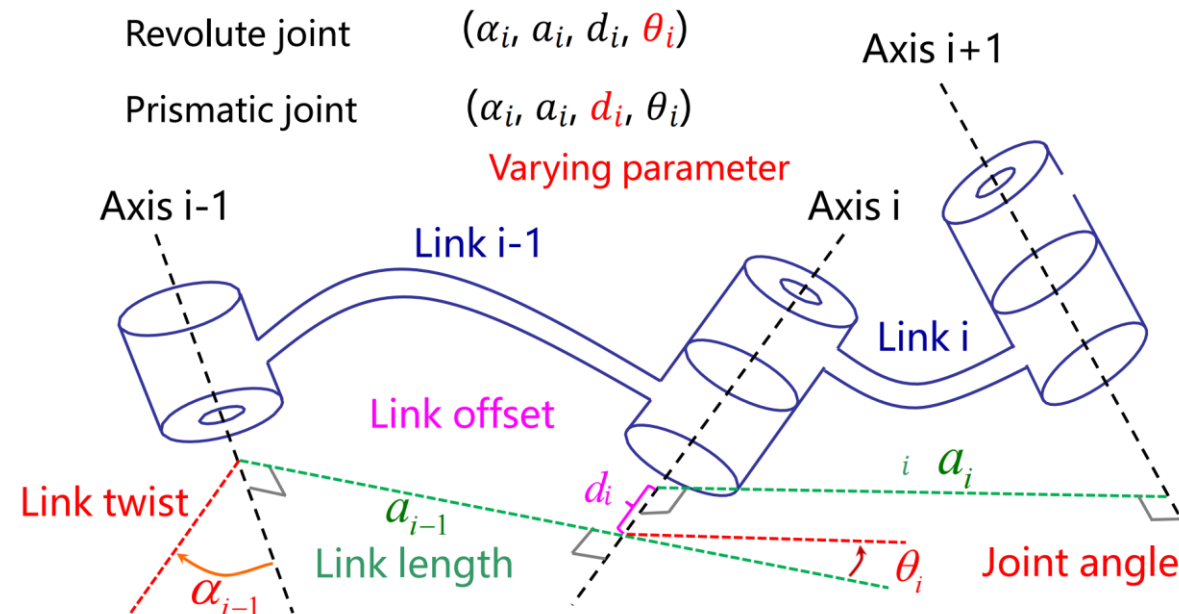
Link twist α : angle (relative location) between two axis.



2. Geometric Description of the Manipulator

❑ **Intermediate Links in the Chain:** In case of multiple links, neighboring links have a common joint axis between them, two more parameters:

- **link offset d_i :** parameter of interconnection, the distance along this common axis from one link to the next.
- **Joint angle θ_i :** describes the amount of rotation about this common axis between one link and its neighbor.



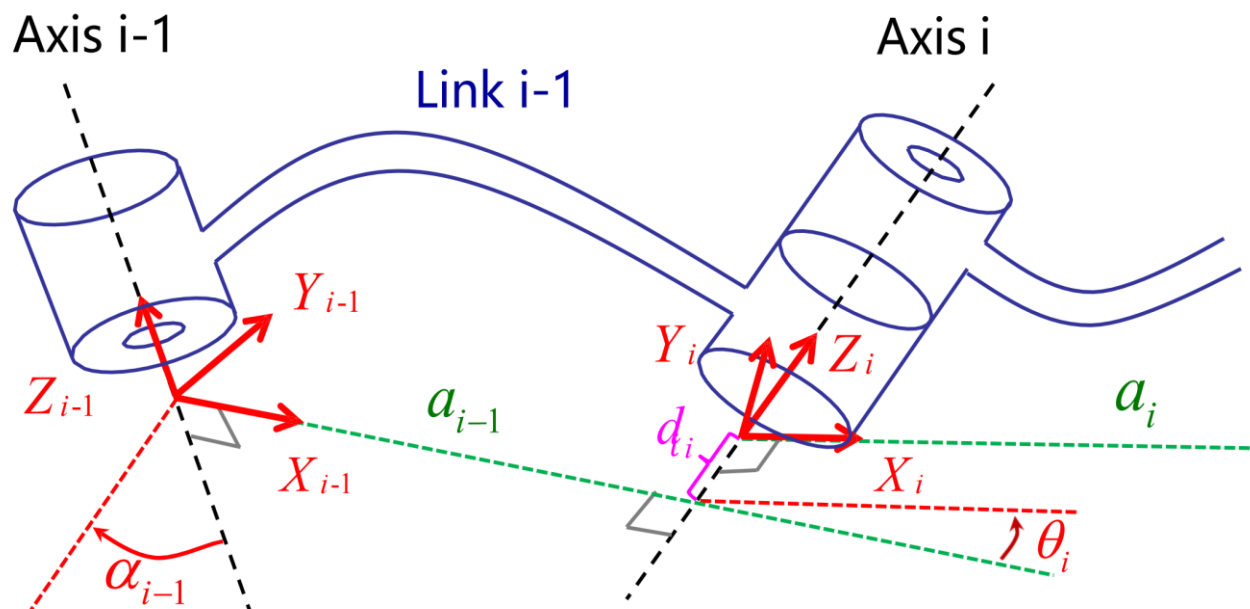
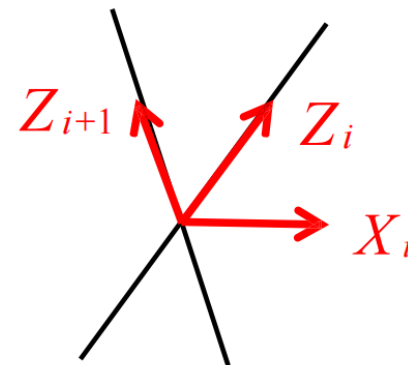
Definition of mechanisms by means of these four quantities is called the **Denavit–Hartenberg (D-H) notation**:

- For a revolute joint: θ_i is called the **joint variable**, and the other three quantities would be fixed **link parameters**.
- For prismatic joints: d_i is the **joint variable**, and the other three quantities are fixed **link parameters**.

3. Definition of the Frame of Link

□ Frame for Link $\{i\}$

- Z_i Direction of rotation or movement of axis (both rotation and linear)
- X_i Along the direction of a_i (if $a_i \neq 0$)
Vertical to both Z_i and Z_{i+1} (if $a_i = 0$)
- Y_i Vertical to both X_i and Z_i , according to the right-hand rule



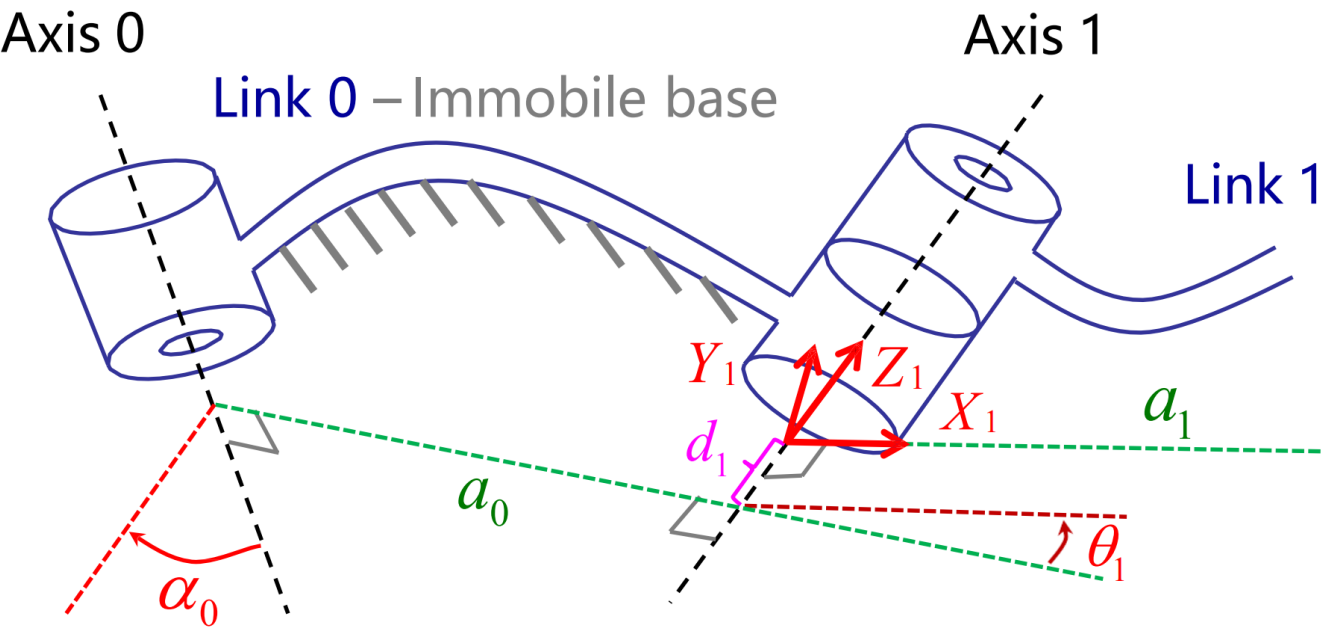
3. Definition of the Frame of Link

□ Frame for Link {0}

Frame {0} coincides with frame {1} $a_0 = 0$ $\alpha_0 = 0$

Revolute joint θ_1 is arbitrary $d_1 = 0$

Prismatic joint d_1 is arbitrary $\theta_1 = 0$



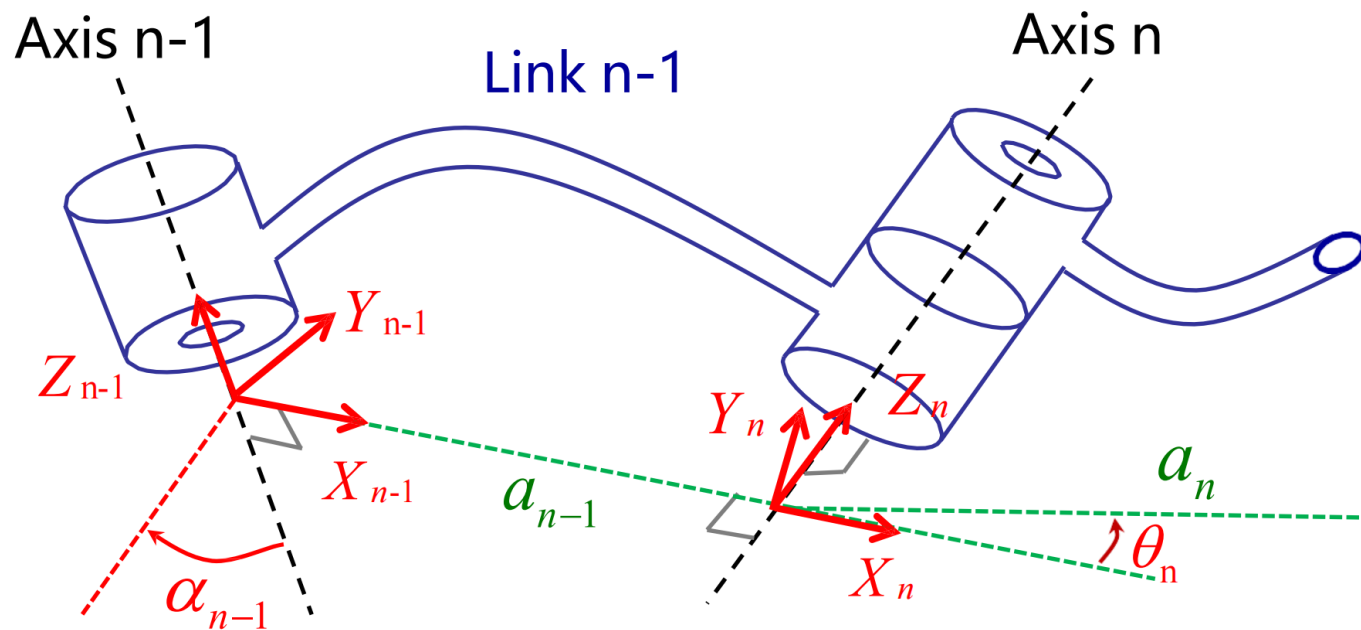
3. Definition of the Frame of Link

□ Frame of the last Link {n}

Take the same direction as X_{n-1} $a_n = 0$ $\alpha_n = 0$

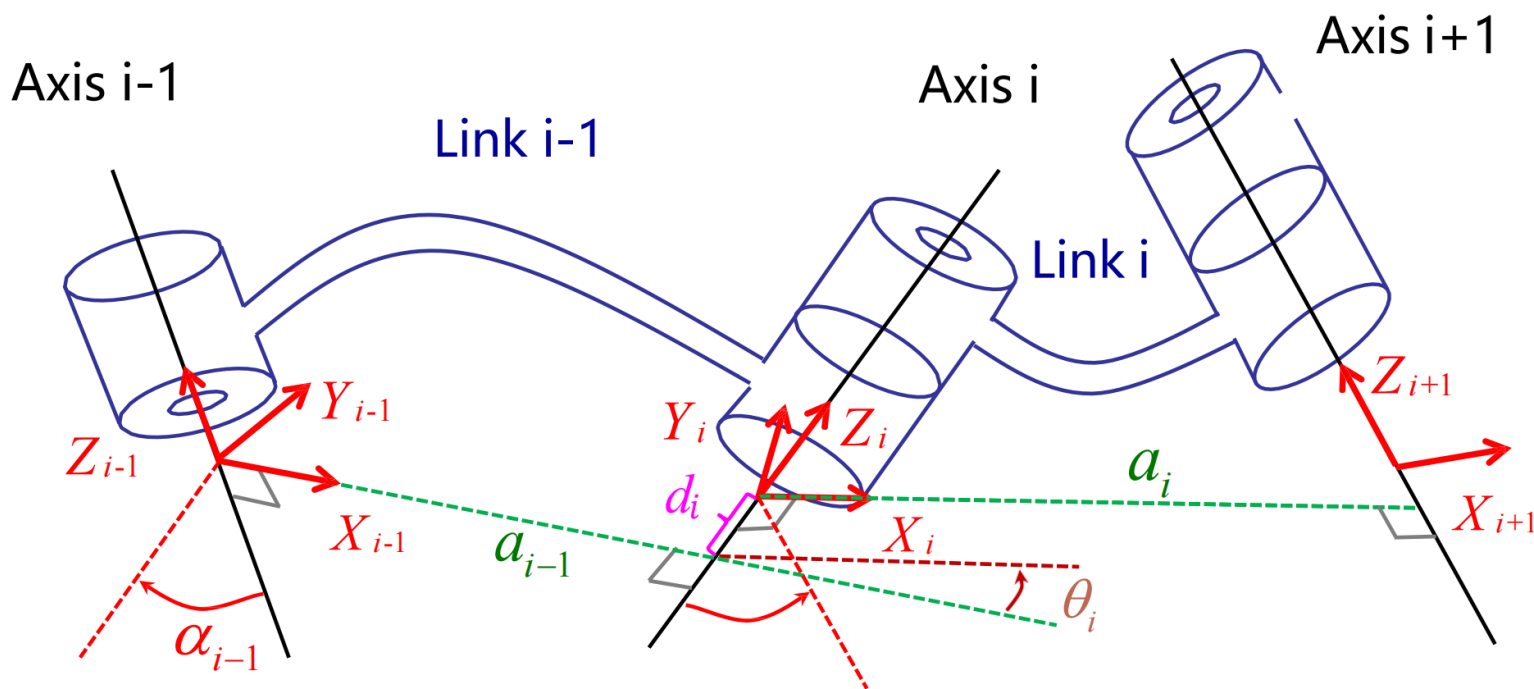
Revolute joint θ_n is arbitrary $d_n = 0$

Prismatic joint d_n is arbitrary $\theta_n = 0$



4. Manipulator Forward Kinematics

- α_{i-1} : the angle between Z_{i-1} and Z_i measured around X_{i-1}
- a_{i-1} : distance between Z_{i-1} and Z_i along X_{i-1} direction ($a_i > 0$)
- θ_i : the angle between X_{i-1} and X_i measured around Z_i
- d_i : distance between X_{i-1} and X_i along Z_i direction



4. Manipulator Forward Kinematics

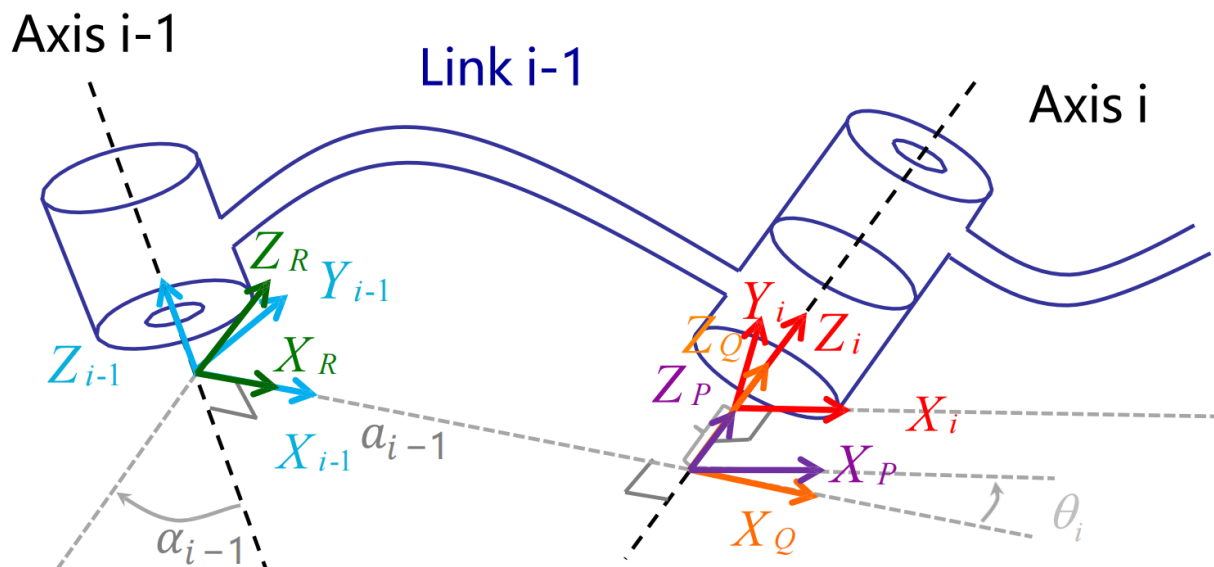
Construct the transform that defines frame $\{i\}$ relative to the frame $\{i-1\}$:

$${}^{i-1}P = {}^{i-1}T_i P$$

$${}^{i-1}P = {}^{i-1}T_R^R T_Q^Q T_P^P T_i^i P$$

$${}^{i-1}T_i = {}^{i-1}T_R^R T_Q^Q T_P^P T_i^i$$

$$= T_{\hat{X}_{i-1}}(\alpha_{i-1}) T_{\hat{X}_R}(a_{i-1}) T_{\hat{Z}_Q}(\theta_i) T_{\hat{Z}_P}(d_i)$$



For any *given* robot, this transformation will be a **function of only one variable**, the other three parameters being fixed by mechanical design.



4. Manipulator Forward Kinematics

□ Expand the transformation

$$\begin{aligned} {}^{i-1}_iT &= T_{\hat{X}_{i-1}}(\alpha_{i-1}) T_{\hat{X}_R}(a_{i-1}) T_{\hat{Z}_Q}(\theta_i) T_{\hat{Z}_P}(d_i) \\ &= \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

□ Successive linear transformations

$${}^0_nT = {}^0_1T {}^1_2T {}^2_3T \dots {}^{n-2}_{n-1}T {}^{n-1}_nT$$

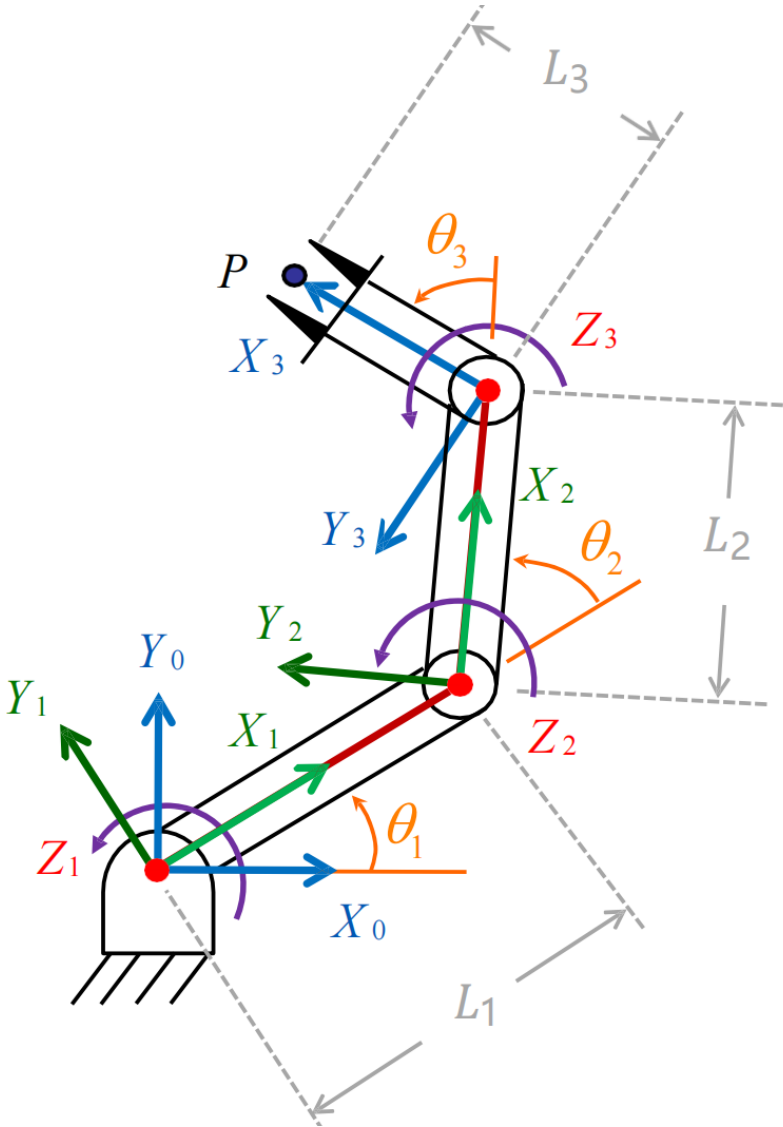
The geometric relationship of frame $\{n\}$ with respect to frame $\{0\}$ is clearly defined: the vectors defined under frame $\{n\}$ can be transferred back to that in frame $\{0\}$.

4. Manipulator Forward Kinematics - Example 1: RRR manipulator

Three-link planar robotic arm

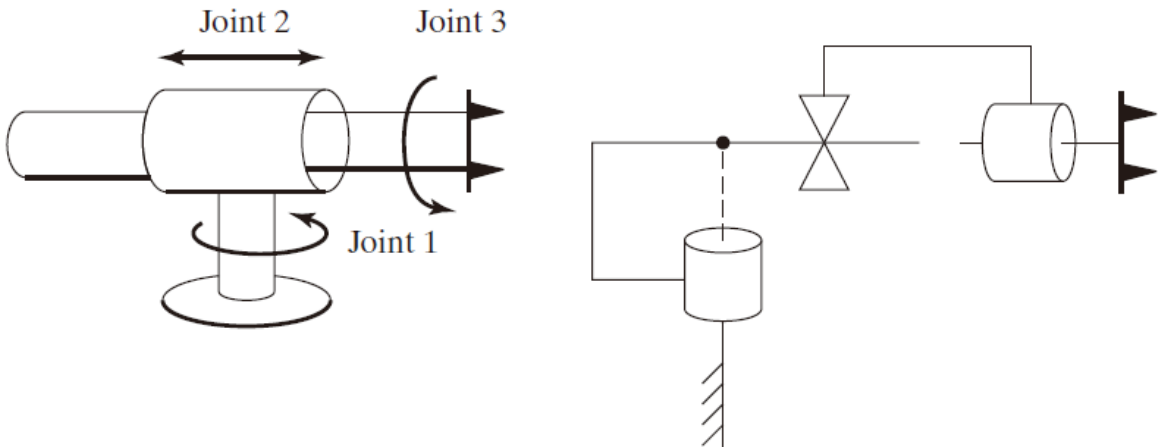
D-H parameters

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	L_1	0	θ_2
3	0	L_2	0	θ_3



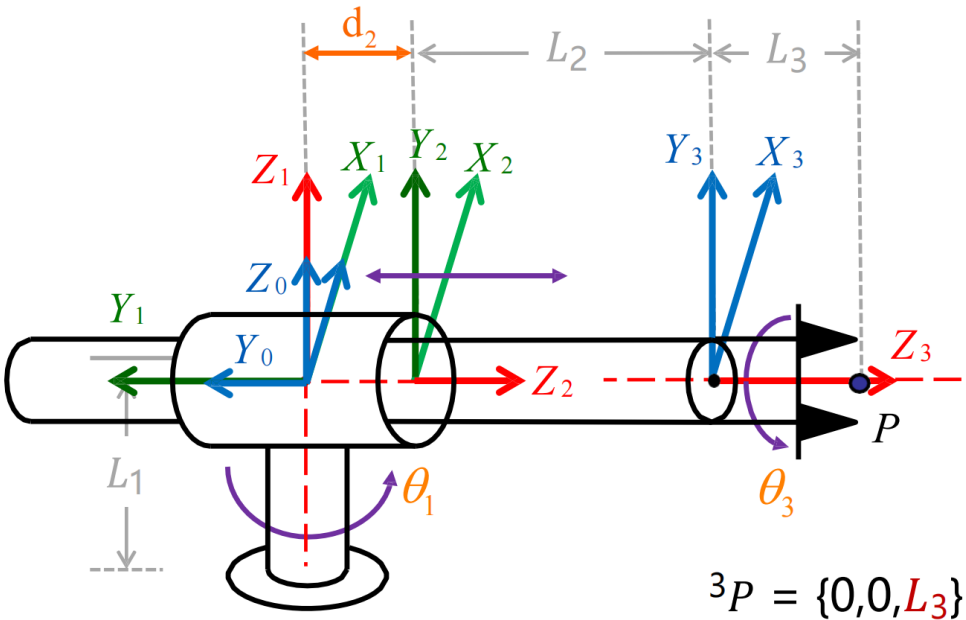
4. Manipulator Forward Kinematics - Example 2: RPR manipulator

- Manipulator having three degrees of freedom, among which one is prismatic joint



D-H parameters

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	90°	0	d_2	0
3	0	0	L_2	θ_3

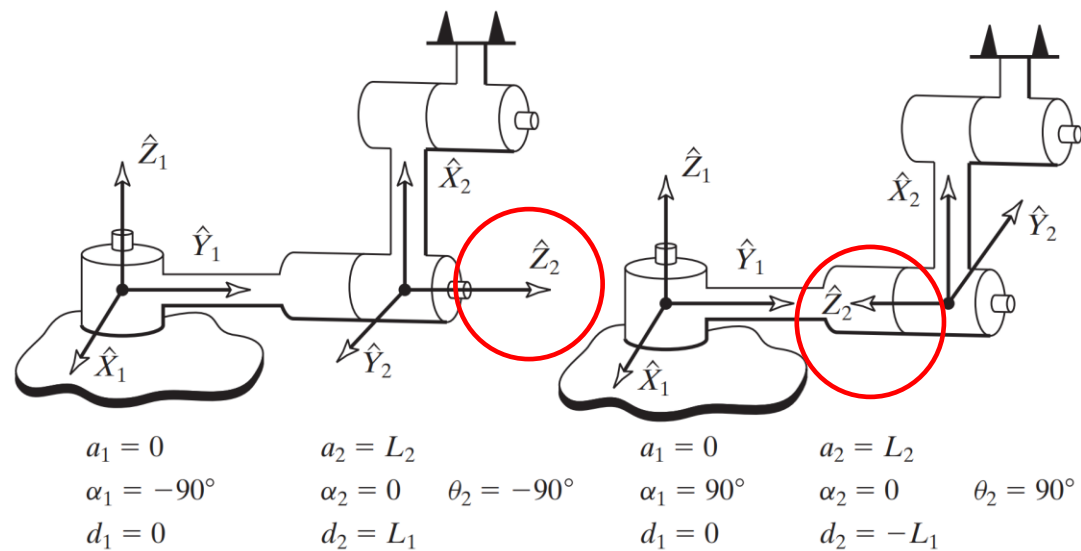


4. Manipulator Forward Kinematics - Example 3: RRR manipulator

When $a_1 = 0$

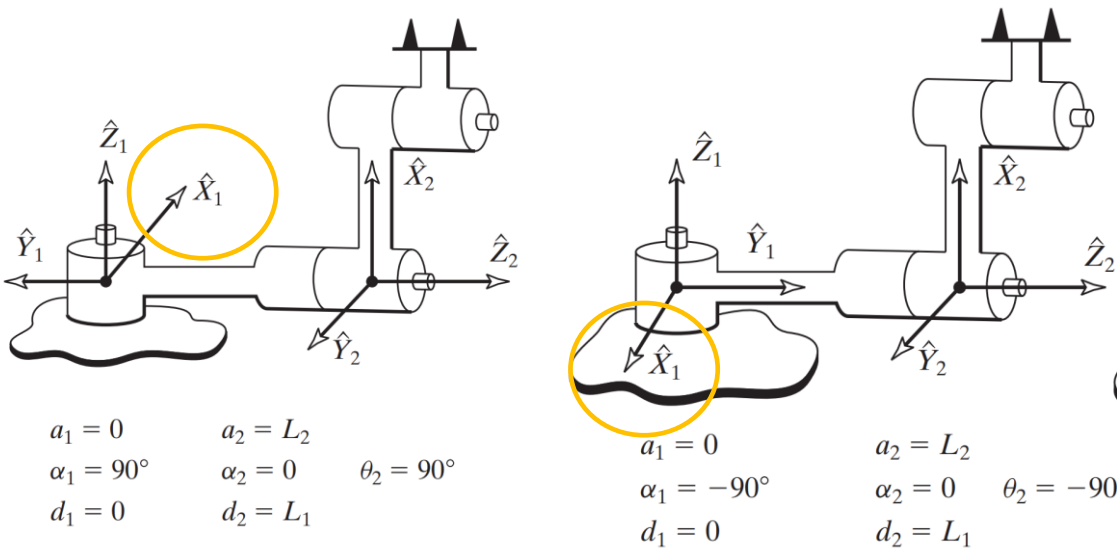
Z_1 and Z_2 intersect

Two options for Z_2



Two possible frame assignments

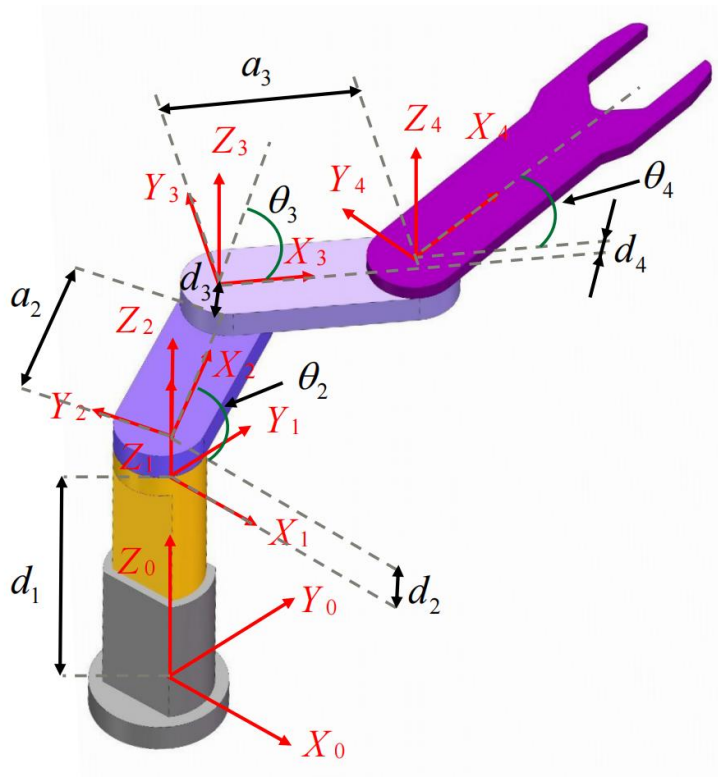
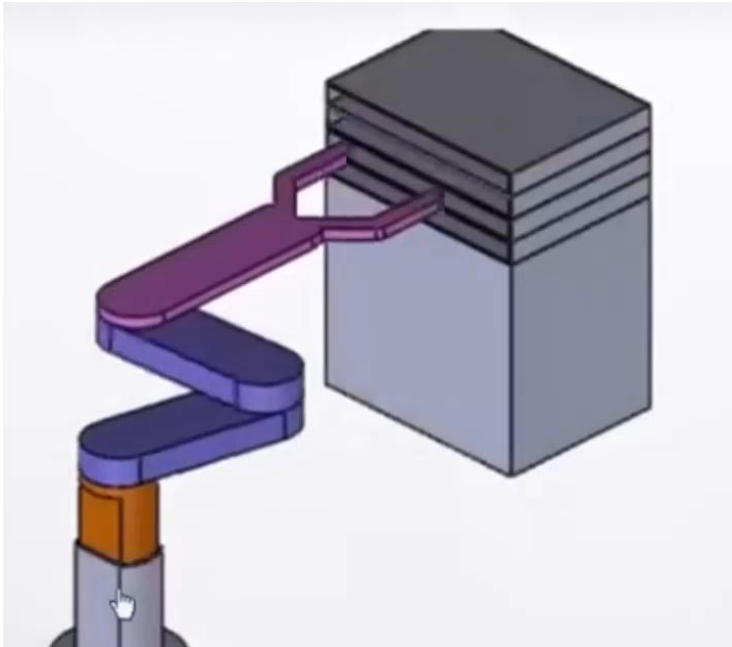
Two options for X_1



Two more possible frame assignments

4. Manipulator Forward Kinematics - Example 4: PRRR manipulator

□ Wafer Robot



$d_1, \theta_2, \theta_3, \theta_4$ are driving joints

$$\theta_1 = 0 (X_0 // X_1)$$

$\alpha_0, \alpha_1, \alpha_2, \alpha_3 = 0$ All axes are parallel

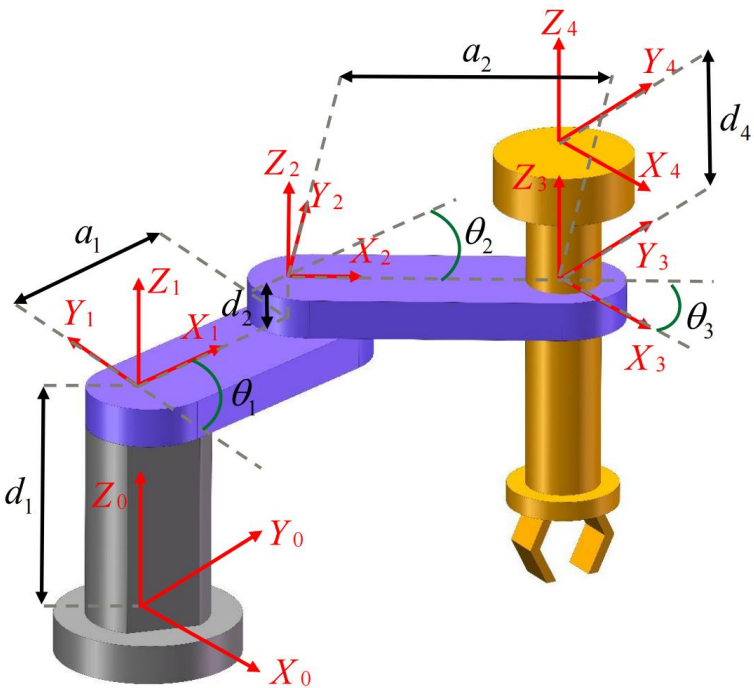
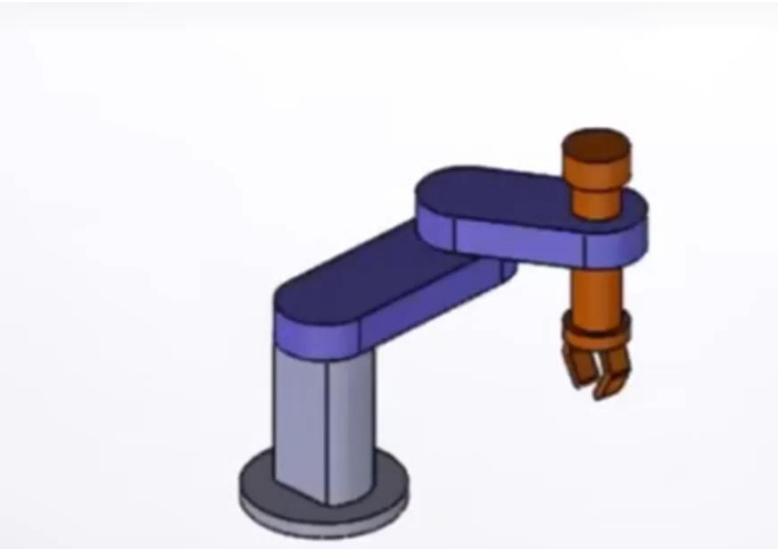
$$a_0, a_1 = 0$$

D-H parameters

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	d_1	0
2	0	0	d_2	θ_2
3	0	a_2	d_3	θ_3
4	0	a_3	d_4	θ_4

4. Manipulator Forward Kinematics - Example 5: RRRP manipulator

□ SCARA robot



$\theta_1, \theta_2, \theta_3, d_4$ are driving joints

$$\theta_1 = 0 (X_0 // X_1)$$

$\alpha_0, \alpha_1, \alpha_2, \alpha_3 = 0$ All axes are parallel

$$a_0, a_3 = 0$$

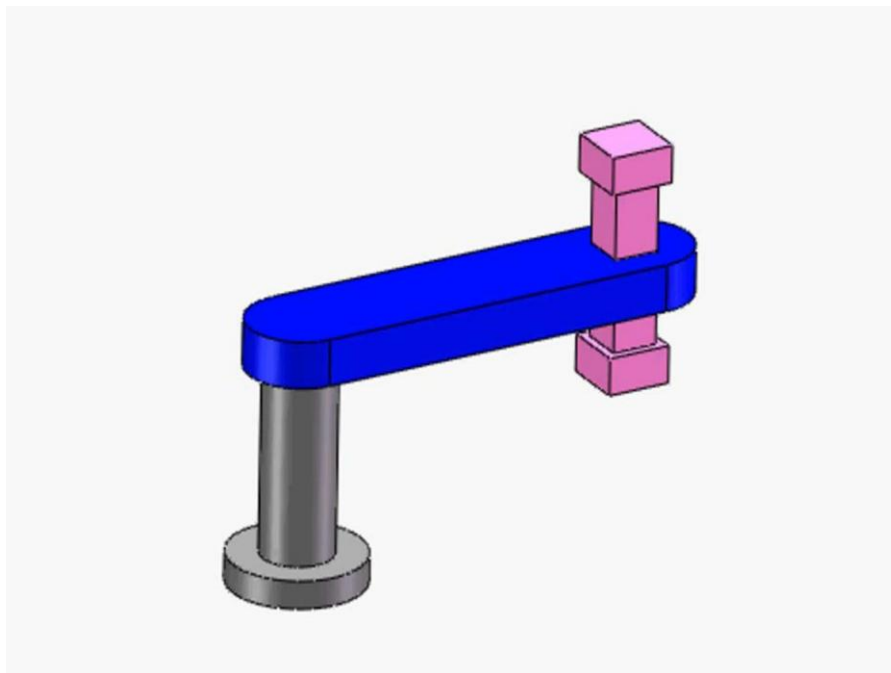
D-H parameters

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	d_1	θ_1
2	0	a_1	d_2	θ_2
3	0	a_2	0	θ_3
4	0	0	d_4	0



4. Manipulator Forward Kinematics - Example 6: RP manipulator

- In class quiz - the manipulator below consists of a revolute joint and a prismatic joint. Among all the DH parameters (a_{i-1} α_{i-1} d_i θ_i), which two parameters are the driving joints?



(A) α_1, a_2

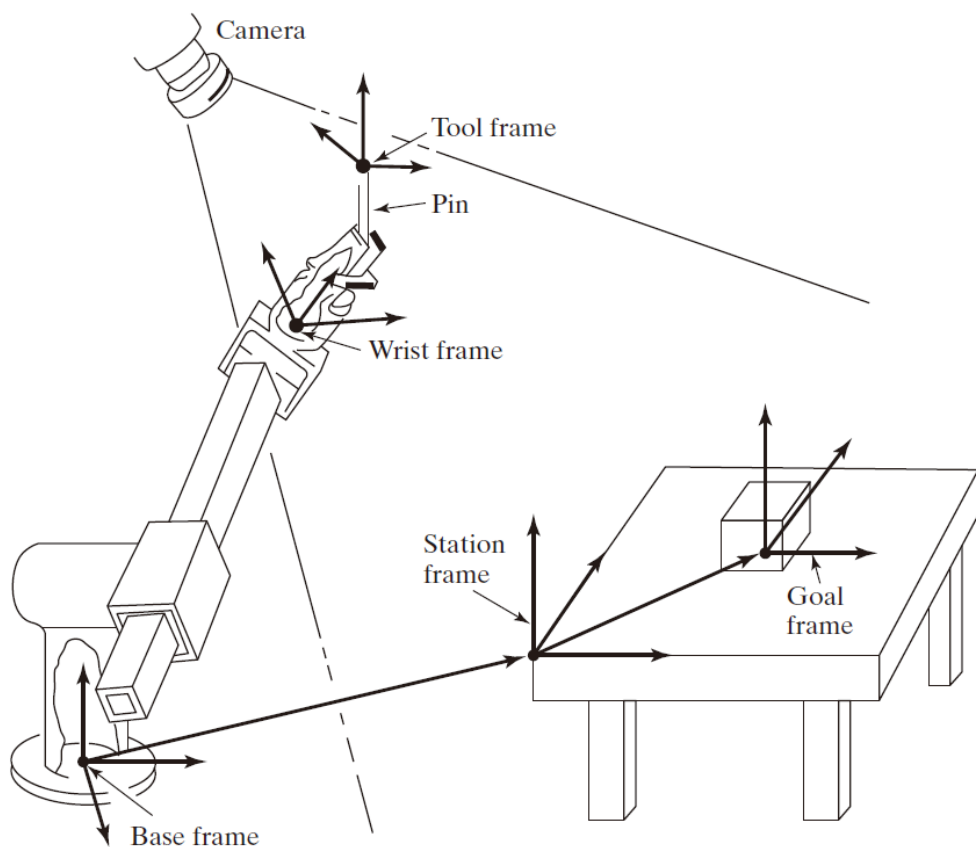
(B) α_1, d_2

(C) θ_1, a_2

(D) θ_1, d_2

5. Frames with Standard Names

- It will be helpful if we assign specific names and locations to certain “standard” frames associated with a robot and its workspace.



Example of the assignment of standard frames

The Base Frame $\{B\}$: the base of the manipulator

The Station Frame $\{S\}$: a task-relevant location

The Wrist Frame $\{W\}$: is affixed to the last link of the manipulator; very often, $\{W\}$ has its origin fixed at a point called the *wrist* of the manipulator, moves with the last link of the manipulator

The Tool Frame $\{T\}$: is affixed to the end of any tool the robot happens to be holding

The Goal Frame $\{G\}$: a description of the location to which the robot is to move the tool

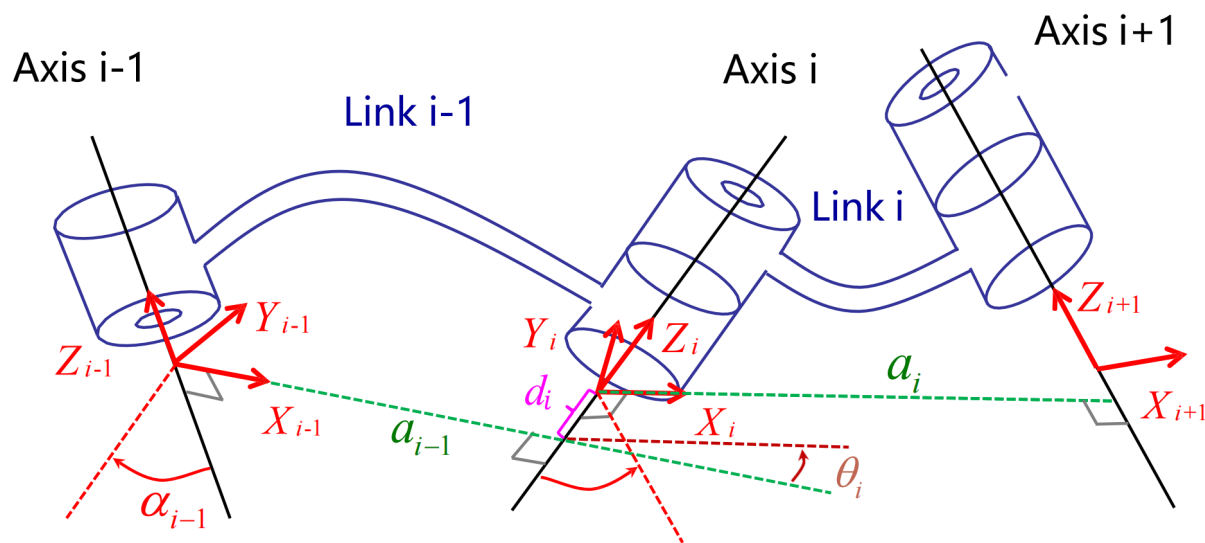
To pick up, we wish to calculate the value of the tool frame $\{T\}$, relative to the station frame, $\{S\}$

$${}^S_T = {}^B_T^{-1} {}^B_W {}^W_T$$

Tool -> Wrist -> Base -> Station

6. Revisit of Denavit-Hartenberg (Modified DH)

- α_{i-1} : around X_{i-1} direction, the angle between Z_{i-1} and Z_i
- a_{i-1} : distance between Z_{i-1} and Z_i along X_{i-1} direction ($a_i > 0$)
- θ_i : around Z_i direction, the angle between X_{i-1} and X_i
- d_i : distance between X_{i-1} and X_i along Z_i direction

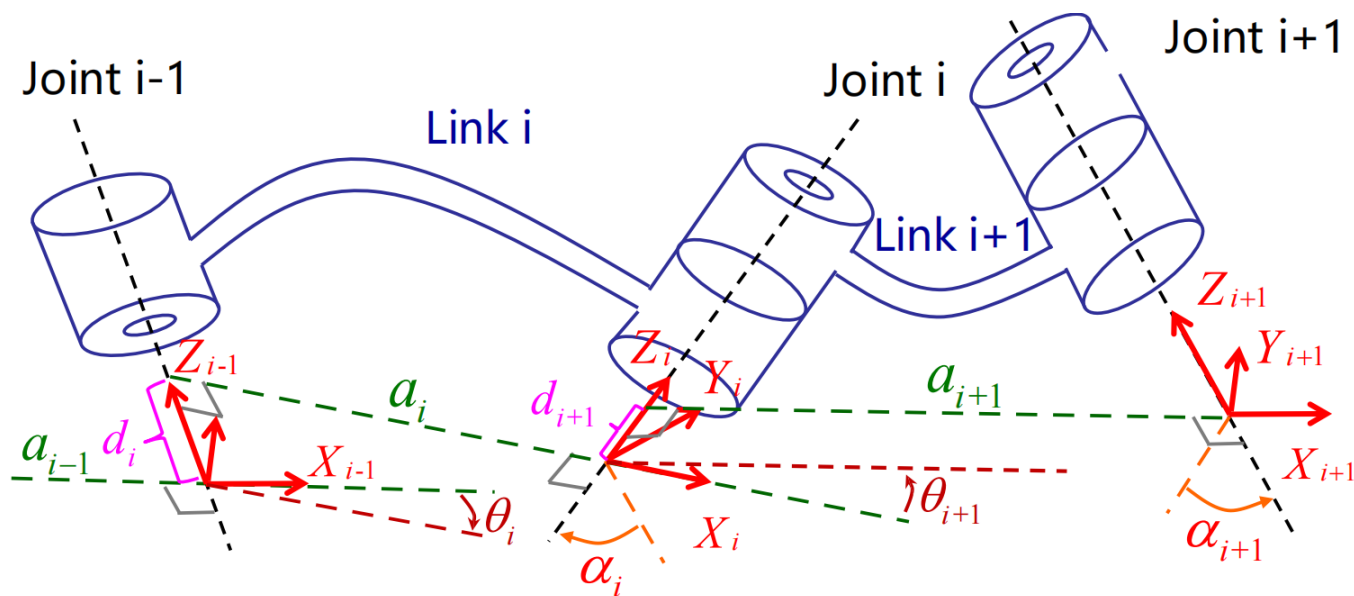


$$\begin{aligned}
 {}^{i-1}_i T &= {}^{i-1}_R T {}^R_Q T {}^Q_P T {}^P_i T \\
 &= T_{X_{i-1}}(\alpha_{i-1}) T_{X_R}(a_{i-1}) T_{Z_Q}(\theta_i) T_{Z_P}(d_i) \\
 &= \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1} d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1} d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$



7. Denavit-Hartenberg (Standard)

- θ_i : around Z_{i-1} direction, the angle between X_{i-1} and X_i
- d_i : along Z_{i-1} direction, distance between X_{i-1} and X_i
- a_i : along X_i direction, distance between Z_{i-1} and Z_i ($a_i > 0$)
- α_i : around X_i direction, the angle between Z_{i-1} and Z_i

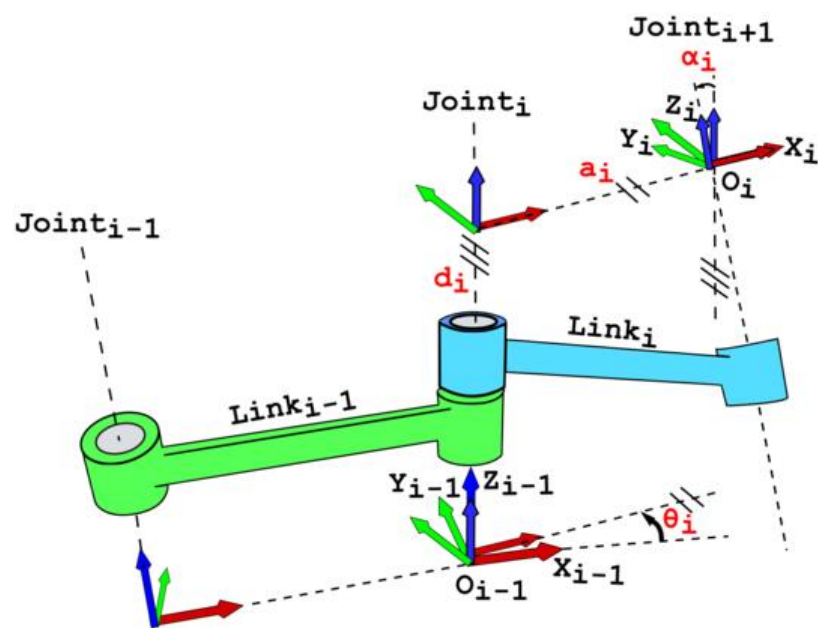


$$\begin{aligned} {}^{i-1}_i T &= {}^{i-1}_R T {}^R_Q T {}^Q_P T {}^P_i T \\ &= T_{Z_{i-1}}(\theta_i) T_{Z_R}(d_i) T_{X_Q}(a_i) T_{X_P}(\alpha_i) \\ &= \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

7. Denavit-Hartenberg (Standard)

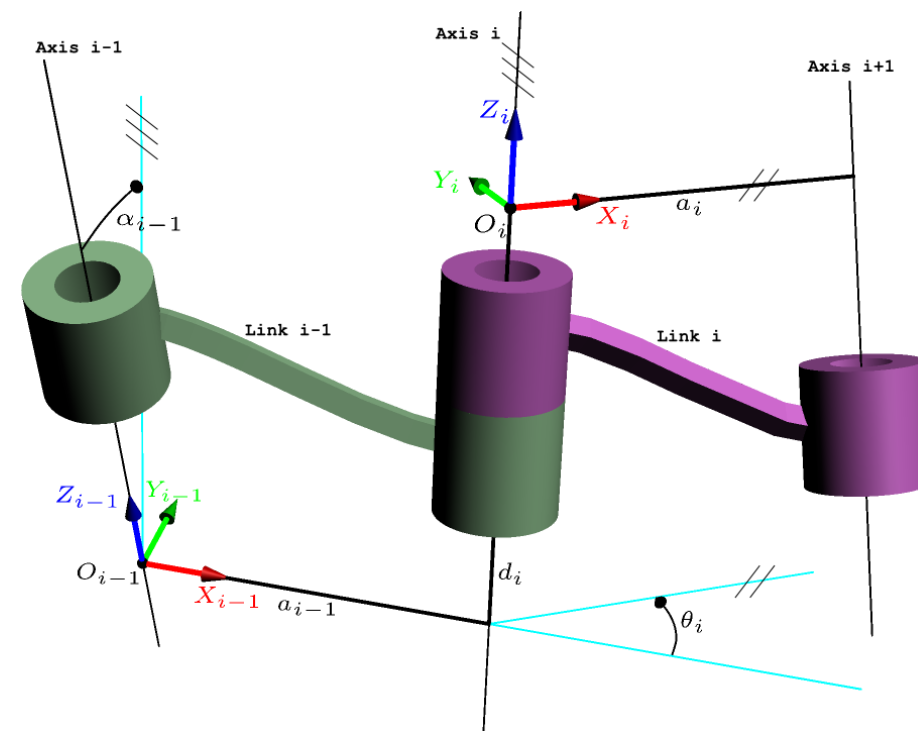
□ The difference between the standard DH parameters and the modified DH parameters:

- (1) the **locations** of the frame attachment to the links.
- (2) the **order** of the performed transformations.



The four parameters of **standard DH** convention are shown in red text.

VS.



The four parameters of **modified DH** convention are shown in black text.

Example 1: RRR Operator: Craig DH (modified DH)

□ Joint axes

□ Common perpendiculars

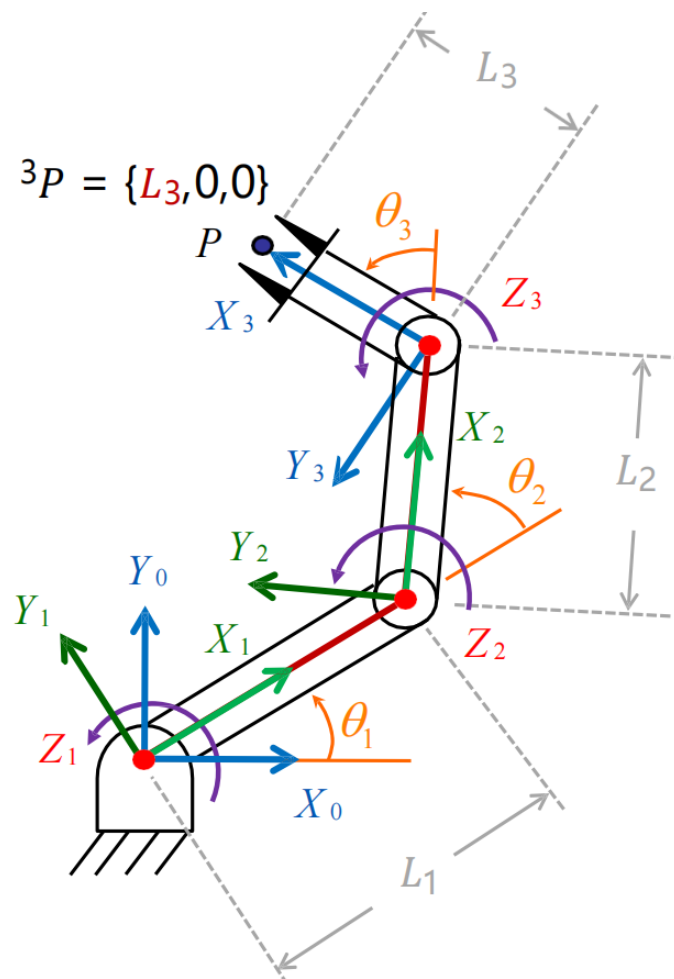
□ Z_i

□ X_i

□ Y_i

□ Frame $\{0\}$ and $\{n\}$

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	L_1	0	θ_2
3	0	L_2	0	θ_3



$${}^0_1T = \begin{pmatrix} \cos[t1] & -\sin[t1] & 0 & 0 \\ \sin[t1] & \cos[t1] & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

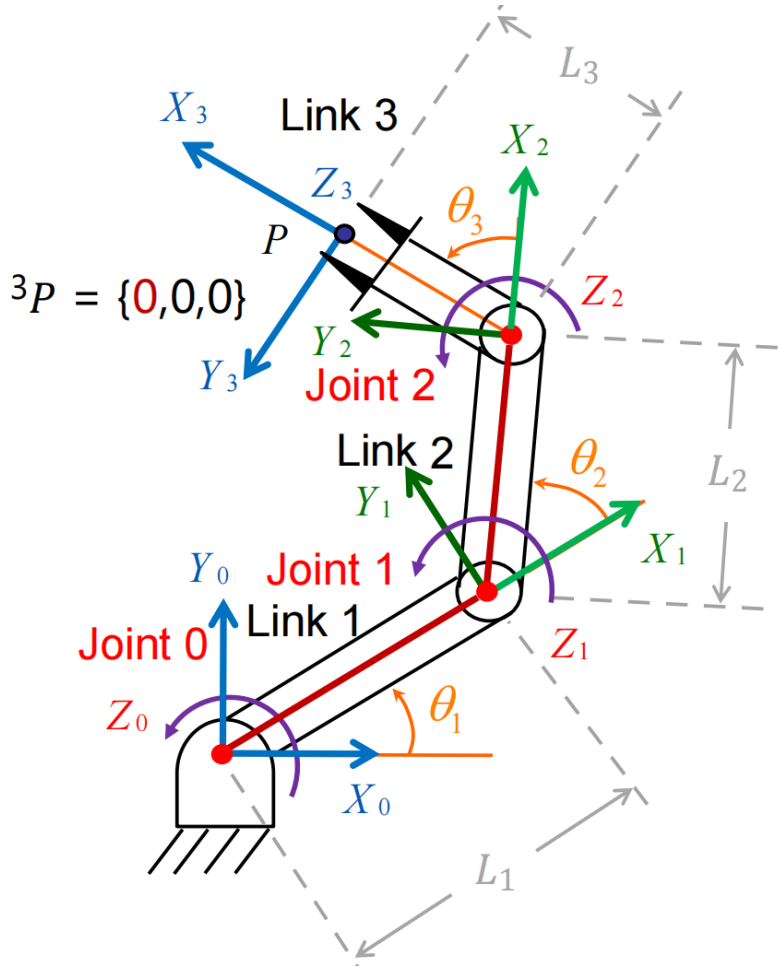
$${}^1_2T = \begin{pmatrix} \cos[t2] & -\sin[t2] & 0 & L1 \\ \sin[t2] & \cos[t2] & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^2_3T = \begin{pmatrix} \cos[t3] & -\sin[t3] & 0 & L2 \\ \sin[t3] & \cos[t3] & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



Example 1: RRR Operator: Standard DH

- Joint axes
- Common perpendiculars
- Z_i
- X_i
- Y_i
- Frame $\{0\}$ and $\{n\}$



$${}^0_1T \begin{pmatrix} \cos[t1] & -\sin[t1] & 0 & L1 \cos[t1] \\ \sin[t1] & \cos[t1] & 0 & L1 \sin[t1] \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^1_2T \begin{pmatrix} \cos[t2] & -\sin[t2] & 0 & L2 \cos[t2] \\ \sin[t2] & \cos[t2] & 0 & L2 \sin[t2] \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^2_3T \begin{pmatrix} \cos[t3] & -\sin[t3] & 0 & L3 \cos[t3] \\ \sin[t3] & \cos[t3] & 0 & L3 \sin[t3] \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

i	α_i	a_i	d_i	θ_i
1	0	L_1	0	θ_1
2	0	L_2	0	θ_2
3	0	L_3	0	θ_3



Example 1: RRR Operator

□ Craig

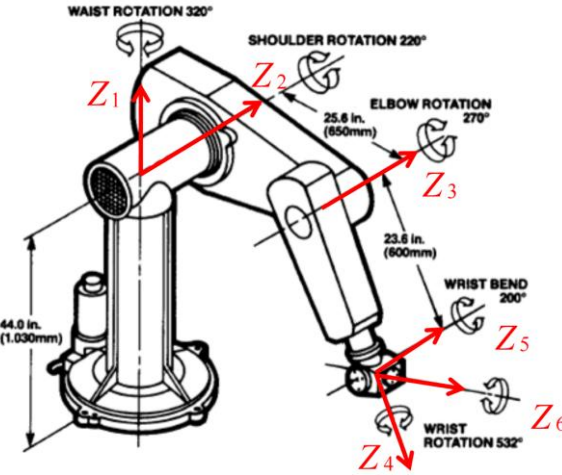
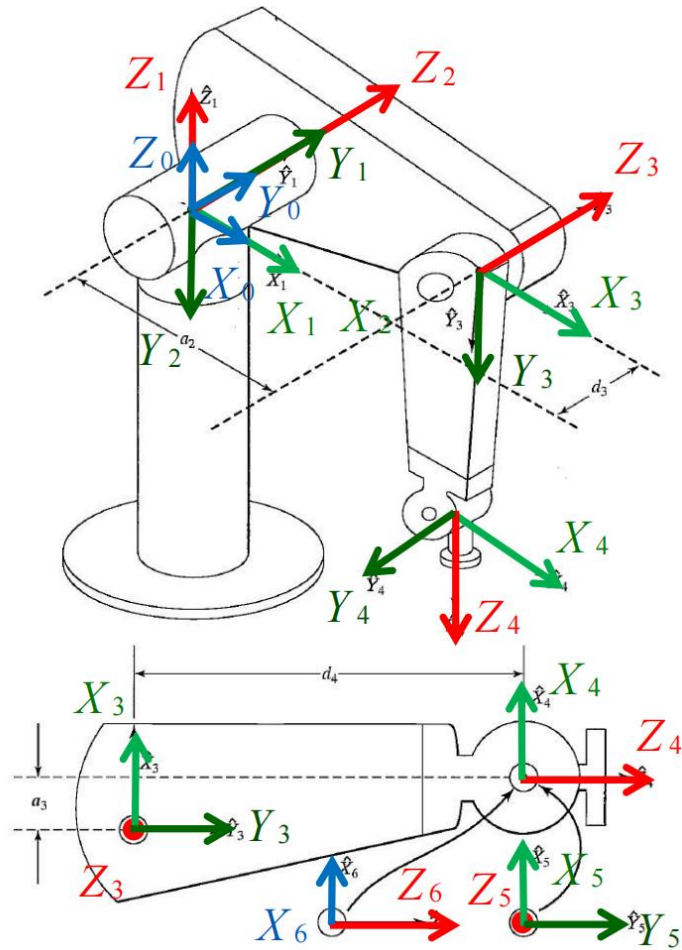
$${}^0_3T \begin{pmatrix} \cos[t1+t2+t3] & -\sin[t1+t2+t3] & 0 & L1\cos[t1]+L2\cos[t1+t2] \\ \sin[t1+t2+t3] & \cos[t1+t2+t3] & 0 & L1\sin[t1]+L2\sin[t1+t2] \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$
$${}^0_3T \cdot T_{\hat{x}_3}([L_3, 0, 0])$$
$$\begin{pmatrix} \cos[t1+t2+t3] & -\sin[t1+t2+t3] & 0 & L1\cos[t1]+L2\cos[t1+t2]+L3\cos[t1+t2+t3] \\ \sin[t1+t2+t3] & \cos[t1+t2+t3] & 0 & L1\sin[t1]+L2\sin[t1+t2]+L3\sin[t1+t2+t3] \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

□ Standard

$${}^0_3T \begin{pmatrix} \cos[t1+t2+t3] & -\sin[t1+t2+t3] & 0 & L1\cos[t1]+L2\cos[t1+t2]+L3\cos[t1+t2+t3] \\ \sin[t1+t2+t3] & \cos[t1+t2+t3] & 0 & L1\sin[t1]+L2\sin[t1+t2]+L3\sin[t1+t2+t3] \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Example 2: PUMA560

□ Frames (Craig-Modified DH)



i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0°	0	0	θ_1
2	-90°	0	0	θ_2
3	0°	a_2	d_3	θ_3
4	-90°	a_3	d_4	θ_4
5	90°	0	0	θ_5
6	-90°	0	0	θ_6

$${}^0_1T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2T = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s\theta_2 & -c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & a_2 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^3_4T = \begin{bmatrix} c\theta_4 & -s\theta_4 & 0 & a_3 \\ 0 & 0 & 1 & d_4 \\ -s\theta_4 & -c\theta_4 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^4_5T = \begin{bmatrix} c\theta_5 & -s\theta_5 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s\theta_5 & c\theta_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^5_6T = \begin{bmatrix} c\theta_6 & -s\theta_6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s\theta_6 & -c\theta_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Example 2: PUMA560

□ Combining transformation matrix

$${}^4_6T = {}^4_5T {}^5_6T = \begin{bmatrix} c_5c_6 & -c_5s_6 & -s_5 & 0 \\ s_6 & c_6 & 0 & 0 \\ s_5c_6 & -s_5s_6 & c_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^3_6T = {}^3_4T {}^4_6T = \begin{bmatrix} c_4c_5c_6 - s_4s_6 & -c_4c_5s_6 - s_4c_6 & -c_4s_5 & a_3 \\ s_5c_6 & -s_5s_6 & c_5 & d_4 \\ -s_4c_5c_6 - c_4s_6 & s_4c_5s_6 - c_4c_6 & s_4s_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_3T = {}^1_2T {}^2_3T = \begin{bmatrix} c_{23} & -s_{23} & 0 & a_2c_2 \\ 0 & 0 & 1 & d_3 \\ -s_{23} & -c_{23} & 0 & -a_2s_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$${}^1_6T = {}^1_3T {}^3_6T = \begin{bmatrix} {}^1r_{11} & {}^1r_{12} & {}^1r_{13} & {}^1p_x \\ {}^1r_{21} & {}^1r_{22} & {}^1r_{23} & {}^1p_y \\ {}^1r_{31} & {}^1r_{32} & {}^1r_{33} & {}^1p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1r_{11} = c_{23}[c_4c_5c_6 - s_4s_6] - s_{23}s_5s_6$$

$${}^1r_{21} = -s_4c_5c_6 - c_4s_6$$

$${}^1r_{31} = -s_{23}[c_4c_5c_6 - s_4s_6] - c_{23}s_5c_6$$

$${}^1r_{12} = -c_{23}[c_4c_5s_6 + s_4s_6] + s_{23}s_5s_6$$

$${}^1r_{22} = s_4c_5s_6 - c_4c_6$$

$${}^1r_{32} = s_{23}[c_4c_5s_6 + s_4s_6] + c_{23}s_5s_6$$

$${}^1r_{13} = -c_{23}c_4s_5 - s_{23}c_5$$

$${}^1r_{23} = s_4s_5$$

$${}^1r_{33} = s_{23}c_4s_5 - c_{23}c_5$$

$${}^1p_x = a_2c_2 + a_3c_{23} - d_4s_{23}$$

$${}^1p_y = d_3$$

$${}^1p_z = -a_3s_{23} - a_2s_2 - d_4c_{23}$$



Summary

- ❑ Construction the frame of link, and the criteria for determining the DH parameters.
- ❑ Transformation matrix of link, the standard naming of the coordinate system.
- ❑ Standard DH method vs. modified DH method.

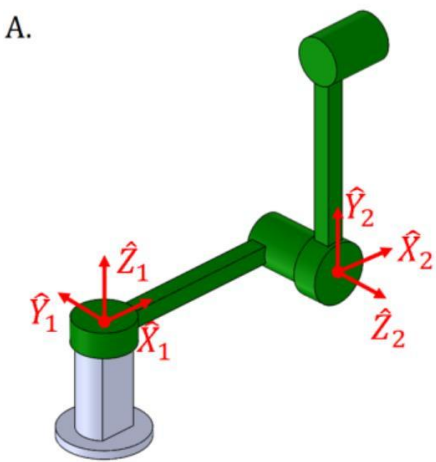
Manipulator Forward Kinematics

- Geometric description of the manipulator
- Definition of frame of link
- Forward kinematics
- Standard naming of frames
- Denavit-Hartenberg expression(Standard)

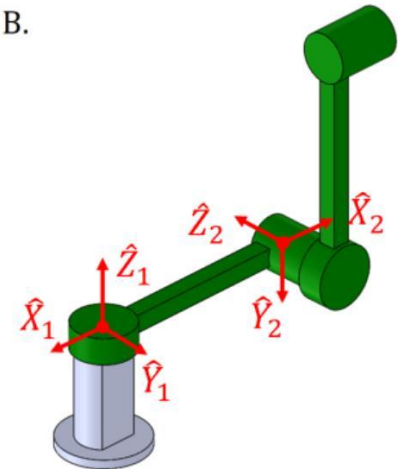
Assignment



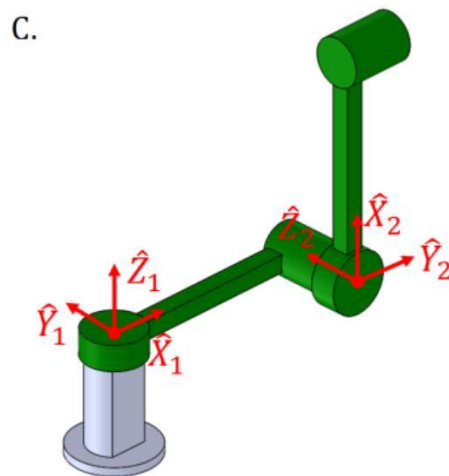
□ Assignment 1 - which of the following is the correct frame assignment on the RR manipulator?



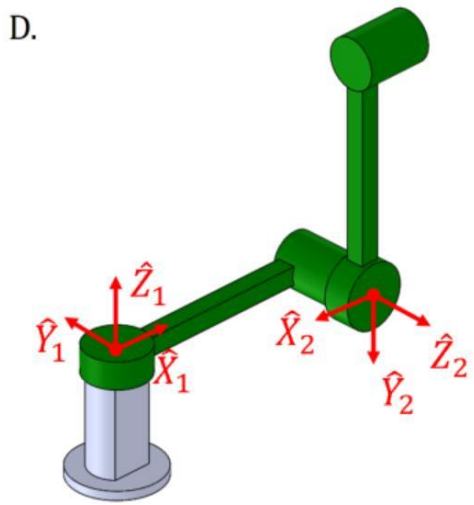
A



B



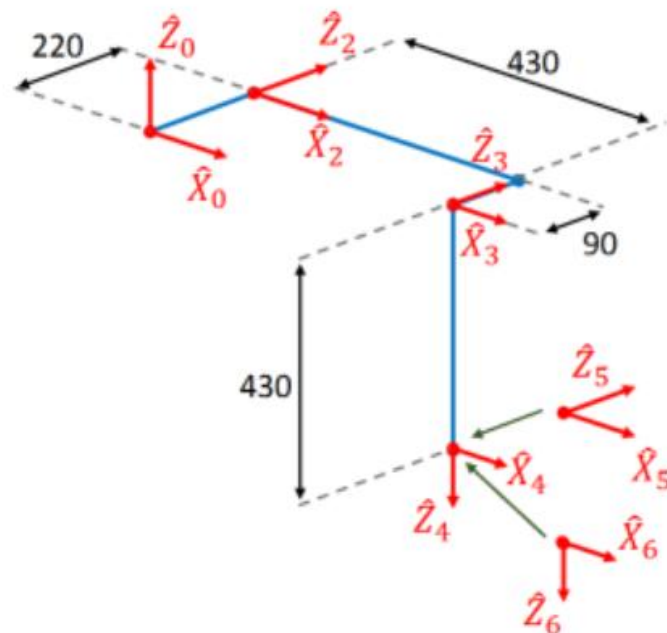
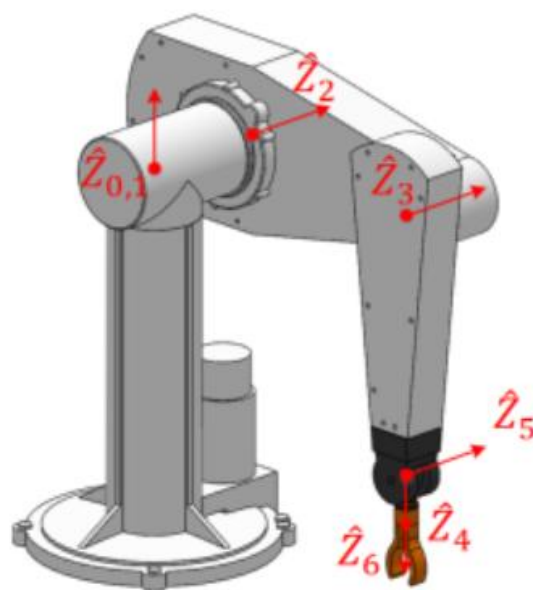
C



D

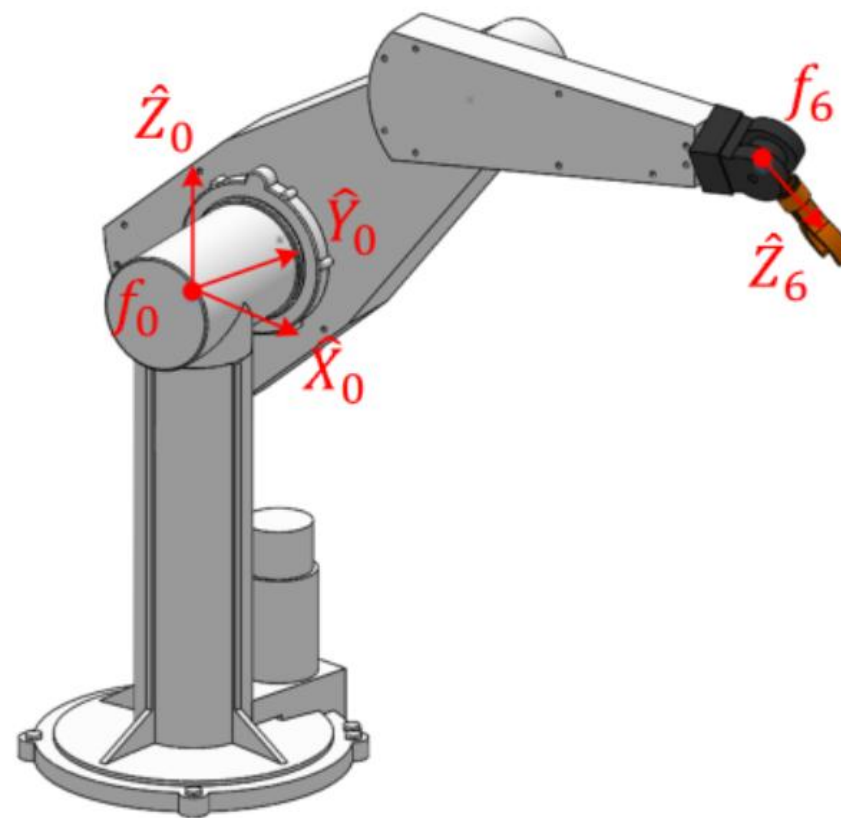
Assignment

- Assignment 2 - the following is a PUMA560 6-axis robotic arm, the arm posture is defined in the figure as 0° in all axes, please refer to the length of each link provided in the figure, the origin of the defined frame $\{0\}$ - $\{6\}$ and the Z-axis, give the DH table.



Assignment

- Assignment 3 - continuing from the previous question, the angles of the six joints are $\theta_1 = 15^\circ$, $\theta_2 = -40^\circ$, $\theta_3 = -50^\circ$, $\theta_4 = 30^\circ$, $\theta_5 = 70^\circ$, $\theta_6 = 25^\circ$.
- What is the homogeneous transformation matrix of frame {2} relative to frame {1} ?
 - What are the coordinates of the origin of frame {4} relative to frame {0}?
 - What is the homogeneous transformation matrix 0T_6 of frame {6} relative to frame {0}?





Q&A